

	$\mathcal{K}_{0^+}^{\#1}$	$\mathcal{K}_{0^+}^{\#1\alpha\beta\chi}$				
$\mathcal{K}_{0^+}^{\#1} \dagger$	$-\frac{3\Upsilon_1 k^2}{2}$	0				
$\mathcal{K}_{0^+}^{\#1} \dagger^{\alpha\beta\chi}$	0	0	$\mathcal{K}_{1^+}^{\#1\alpha\beta}$	$\mathcal{K}_{1^+}^{\#2\alpha\beta}$	$\mathcal{K}_{1^+}^{\#1\alpha}$	$\mathcal{K}_{1^+}^{\#2\alpha}$
	$\mathcal{K}_{1^+}^{\#1} \dagger^{\alpha\beta}$	$-k^2 \frac{{}^{(4)}\kappa_3}{\sqrt{2}}$	$\frac{k^2}{2} \frac{{}^{(4)}\kappa_4}{\sqrt{2}}$	0	0	
	$\mathcal{K}_{1^+}^{\#2} \dagger^{\alpha\beta}$	$\frac{k^2}{2} \frac{{}^{(4)}\kappa_4}{\sqrt{2}}$	$-\frac{\Upsilon_2 k^2}{2}$	0	0	
	$\mathcal{K}_{1^+}^{\#1} \dagger^\alpha$	0	0	$-k^2 \frac{{}^{(4)}\kappa_2}{\sqrt{2}}$	$-\sqrt{2} k^2 \frac{{}^{(4)}\kappa_2}{\sqrt{2}}$	
	$\mathcal{K}_{1^+}^{\#2} \dagger^\alpha$	0	0	$-\sqrt{2} k^2 \frac{{}^{(4)}\kappa_2}{\sqrt{2}}$	$-2 k^2 \frac{{}^{(4)}\kappa_2}{\sqrt{2}}$	$\mathcal{K}_{2^+}^{\#1\alpha\beta}$ $\mathcal{K}_{2^+}^{\#1\alpha\beta\chi}$
				$\mathcal{K}_{2^+}^{\#1} \dagger^{\alpha\beta}$	0	0
				$\mathcal{K}_{2^+}^{\#1} \dagger^{\alpha\beta\chi}$	0	0

	$\mathcal{J}_{0^+}^{\#1}$	$\mathcal{J}_{0^+}^{\#1\alpha\beta\chi}$						
$\mathcal{J}_{0^+}^{\#1} \dagger$	$\frac{1}{\text{Det}(0^+)}$	0						
$\mathcal{J}_{0^+}^{\#1} \dagger^{\alpha\beta\chi}$	0	0	$\mathcal{J}_{1^+}^{\#1\alpha\beta}$	$\mathcal{J}_{1^+}^{\#2\alpha\beta}$	$\mathcal{J}_{1^+}^{\#1\alpha}$	$\mathcal{J}_{1^+}^{\#2\alpha}$		
	$\mathcal{J}_{1^+}^{\#1} \dagger^{\alpha\beta}$		$-\frac{\Upsilon_2 k^2}{2 \text{Det}(1^+)}$	$-\frac{k^2 \overset{(4)}{\kappa}_4}{2 \sqrt{2} \text{Det}(1^+)}$	0	0		
	$\mathcal{J}_{1^+}^{\#2} \dagger^{\alpha\beta}$		$-\frac{k^2 \overset{(4)}{\kappa}_4}{2 \sqrt{2} \text{Det}(1^+)}$	$-\frac{k^2 \overset{(4)}{\kappa}_3}{\text{Det}(1^+)}$	0	0		
	$\mathcal{J}_{1^+}^{\#1} \dagger^\alpha$		0	0	$-\frac{1}{9 k^2 \overset{(4)}{\kappa}_2}$	$-\frac{\sqrt{2}}{9 k^2 \overset{(4)}{\kappa}_2}$		
	$\mathcal{J}_{1^+}^{\#2} \dagger^\alpha$		0	0	$-\frac{\sqrt{2}}{9 k^2 \overset{(4)}{\kappa}_2}$	$-\frac{2}{9 k^2 \overset{(4)}{\kappa}_2}$	$\mathcal{J}_{2^+}^{\#1\alpha\beta}$	$\mathcal{J}_{2^+}^{\#1\alpha\beta\chi}$
					$\mathcal{J}_{2^+}^{\#1} \dagger^{\alpha\beta}$	0	0	
					$\mathcal{J}_{2^+}^{\#1} \dagger^{\alpha\beta\chi}$	0	0	

Abbreviations used in matrices			
$\Upsilon_1 == \overset{(4)}{\kappa}_1 + \overset{(4)}{\kappa}_2 \&\& \Upsilon_2 == 9 \overset{(4)}{\kappa}_2 - \overset{(4)}{\kappa}_3 - \overset{(4)}{\kappa}_4 \&\&$ $\text{Det}(0^+) == -\frac{3}{2} k^2 (\overset{(4)}{\kappa}_1 + \overset{(4)}{\kappa}_2) \&\& \text{Det}(1^+) == \frac{1}{8} k^4 (36 \overset{(4)}{\kappa}_2 \overset{(4)}{\kappa}_3 - (2 \overset{(4)}{\kappa}_3 + \overset{(4)}{\kappa}_4)^2)$			
Lagrangian			
$\overset{(4)}{\kappa}_1 \partial_\beta \mathcal{K}^\delta_{\chi\delta} \partial^\chi \mathcal{K}^\alpha_{\alpha}{}^\beta + \overset{(4)}{\kappa}_2 \partial_\chi \mathcal{K}^\delta_{\beta\delta} \partial^\chi \mathcal{K}^\alpha_{\alpha}{}^\beta + \overset{(4)}{\kappa}_3 \partial_\beta \mathcal{K}^{\alpha\beta\chi} \partial_\delta \mathcal{K}^\delta_{\alpha\chi} + \overset{(4)}{\kappa}_4 \partial_\alpha \mathcal{K}^{\alpha\beta\chi} \partial_\delta \mathcal{K}^\delta_{\beta\chi} - \overset{(4)}{\kappa}_3 \partial_\beta \mathcal{K}^{\alpha\beta\chi} \partial_\delta \mathcal{K}^\delta_{\chi\alpha} -$ $6 \overset{(4)}{\kappa}_2 \partial^\chi \mathcal{K}^\alpha_{\alpha}{}^\beta \partial_\delta \mathcal{K}^\delta_{\chi\beta} - \frac{9}{2} \overset{(4)}{\kappa}_2 \partial_\alpha \mathcal{K}^{\alpha\beta\chi} \partial_\delta \mathcal{K}^\delta_{\beta\chi} + \frac{1}{2} \overset{(4)}{\kappa}_3 \partial_\alpha \mathcal{K}^{\alpha\beta\chi} \partial_\delta \mathcal{K}^\delta_{\beta\chi} + \frac{1}{2} \overset{(4)}{\kappa}_4 \partial_\alpha \mathcal{K}^{\alpha\beta\chi} \partial_\delta \mathcal{K}^\delta_{\beta\chi}$			
Added source term(s):	$\mathcal{K}^{\alpha\beta\chi} \mathcal{J}_{\alpha\beta\chi}$		
Source constraint(s)	# constraint(s)	Covariant form	
$\mathcal{J}_0^{\#1\alpha\beta\chi} == 0$	1	$\partial_\delta \partial^\alpha \mathcal{J}^{\beta\chi\delta} + \partial_\delta \partial^\alpha \mathcal{J}^{\delta\beta\chi} + \partial_\delta \partial^\beta \mathcal{J}^{\chi\alpha\delta} + \partial_\delta \partial^\alpha \mathcal{J}^{\alpha\beta\delta} + \partial_\delta \partial^\chi \mathcal{J}^{\delta\alpha\beta} + \partial_\delta \partial^\delta \mathcal{J}^{\beta\alpha\chi} ==$ $\partial_\delta \partial^\alpha \mathcal{J}^{\chi\beta\delta} + \partial_\delta \partial^\beta \mathcal{J}^{\alpha\chi\delta} + \partial_\delta \partial^\beta \mathcal{J}^{\delta\alpha\chi} + \partial_\delta \partial^\chi \mathcal{J}^{\beta\alpha\delta} + \partial_\delta \partial^\delta \mathcal{J}^{\alpha\beta\chi} + \partial_\delta \partial^\delta \mathcal{J}^{\chi\alpha\beta}$	
$\mathcal{J}_1^{\#1\alpha} == \mathcal{J}_1^{\#2\alpha}$	3	$\partial_\chi \partial^\alpha \mathcal{J}^\beta_{\beta}{}^\chi + \partial_\chi \partial^\chi \mathcal{J}^{\beta\alpha}_{\beta} == 0$	
$\mathcal{J}_2^{\#1\alpha\beta\chi} == 0$	5	$3 \partial_\epsilon \partial_\delta \partial^\chi \partial^\alpha \mathcal{J}^{\delta\beta\epsilon} + 3 \partial_\epsilon \partial^\epsilon \partial^\chi \partial^\alpha \mathcal{J}^{\delta\beta}_{\delta} + 2 \partial_\epsilon \partial^\epsilon \partial_\delta \partial^\beta \mathcal{J}^{\chi\alpha\delta} + 4 \partial_\epsilon \partial^\epsilon \partial_\delta \partial^\beta \mathcal{J}^{\chi\alpha\delta} + 2 \partial_\epsilon \partial^\epsilon \partial_\delta \partial^\beta \mathcal{J}^{\delta\alpha\chi} + 2 \partial_\epsilon \partial^\epsilon \partial_\delta \partial^\chi \mathcal{J}^{\beta\alpha\delta} +$ $4 \partial_\epsilon \partial^\epsilon \partial_\delta \partial^\chi \mathcal{J}^{\delta\alpha\beta} + 2 \partial_\epsilon \partial^\epsilon \partial_\delta \partial^\delta \mathcal{J}^{\alpha\beta\chi} + 3 \eta^{\beta\chi} \partial_\phi \partial^\phi \partial_\epsilon \partial^\alpha \mathcal{J}^{\delta}_{\delta}{}^\epsilon + 3 \eta^{\alpha\chi} \partial_\phi \partial^\phi \partial_\epsilon \partial_\delta \mathcal{J}^{\delta\beta\epsilon} + 3 \eta^{\beta\chi} \partial_\phi \partial^\phi \partial_\epsilon \partial^\epsilon \mathcal{J}^{\delta\alpha}_{\delta} ==$ $3 \partial_\epsilon \partial_\delta \partial^\chi \partial^\beta \mathcal{J}^{\delta\alpha\epsilon} + 3 \partial_\epsilon \partial^\epsilon \partial^\chi \partial^\beta \mathcal{J}^{\delta\alpha}_{\delta} + 2 \partial_\epsilon \partial^\epsilon \partial_\delta \partial^\alpha \mathcal{J}^{\beta\chi\delta} + 4 \partial_\epsilon \partial^\epsilon \partial_\delta \partial^\alpha \mathcal{J}^{\chi\beta\delta} + 2 \partial_\epsilon \partial^\epsilon \partial_\delta \partial^\alpha \mathcal{J}^{\delta\beta\chi} + 2 \partial_\epsilon \partial^\epsilon \partial_\delta \partial^\chi \mathcal{J}^{\alpha\beta\delta} +$ $2 \partial_\epsilon \partial^\epsilon \partial_\delta \partial^\delta \mathcal{J}^{\beta\alpha\chi} + 4 \partial_\epsilon \partial^\epsilon \partial_\delta \partial^\delta \mathcal{J}^{\chi\alpha\beta} + 3 \eta^{\alpha\chi} \partial_\phi \partial^\phi \partial_\epsilon \partial^\beta \mathcal{J}^{\delta}_{\delta}{}^\epsilon + 3 \eta^{\beta\chi} \partial_\phi \partial^\phi \partial_\epsilon \partial_\delta \mathcal{J}^{\delta\alpha\epsilon} + 3 \eta^{\alpha\chi} \partial_\phi \partial^\phi \partial_\epsilon \partial^\epsilon \mathcal{J}^{\delta\beta}_{\delta}$	
$\mathcal{J}_{2^+}^{\#1\alpha\beta} == 0$	5	$3 \partial_\delta \partial_\chi \partial^\alpha \mathcal{J}^{\chi\beta\delta} + 3 \partial_\delta \partial_\chi \partial^\beta \mathcal{J}^{\chi\alpha\delta} + 2 \eta^{\alpha\beta} \partial_\epsilon \partial^\epsilon \partial_\delta \mathcal{J}^\chi_{\chi}{}^\delta == 2 \partial_\delta \partial^\beta \partial^\alpha \mathcal{J}^\chi_{\chi}{}^\delta + 3 (\partial_\delta \partial^\delta \partial_\chi \mathcal{J}^{\alpha\beta\chi} + \partial_\delta \partial^\delta \partial_\chi \mathcal{J}^{\beta\alpha\chi})$	
Total # constraint(s):	14		
Resolved pole(s)	# polarization(s)	Square mass	Residue
	1	0	$-\frac{216 \overset{(4)}{\kappa}_2^2 - 24 \overset{(4)}{\kappa}_2 (2 \overset{(4)}{\kappa}_3 + \overset{(4)}{\kappa}_4) + (2 \overset{(4)}{\kappa}_3 + \overset{(4)}{\kappa}_4)^2}{\overset{(4)}{\kappa}_2 (36 \overset{(4)}{\kappa}_2 \overset{(4)}{\kappa}_3 - (2 \overset{(4)}{\kappa}_3 + \overset{(4)}{\kappa}_4)^2)}$
	1	0	$-\frac{48 \overset{(4)}{\kappa}_2^2 - 8 \overset{(4)}{\kappa}_2 (2 \overset{(4)}{\kappa}_3 + \overset{(4)}{\kappa}_4) + (2 \overset{(4)}{\kappa}_3 + \overset{(4)}{\kappa}_4)^2}{\overset{(4)}{\kappa}_2 (36 \overset{(4)}{\kappa}_2 \overset{(4)}{\kappa}_3 - (2 \overset{(4)}{\kappa}_3 + \overset{(4)}{\kappa}_4)^2)}$
Resolved unitarity condition(s):		$(\overset{(4)}{\kappa}_4 < 0 \&\& ((\overset{(4)}{\kappa}_3 < 0 \&\& (\overset{(4)}{\kappa}_2 < \frac{4 \overset{(4)}{\kappa}_3^2 + 4 \overset{(4)}{\kappa}_3 \overset{(4)}{\kappa}_4 + \overset{(4)}{\kappa}_4^2}{36 \overset{(4)}{\kappa}_3} \parallel \overset{(4)}{\kappa}_2 > 0))) \parallel (\overset{(4)}{\kappa}_3 == 0 \&\& \overset{(4)}{\kappa}_2 > 0) \parallel$ $(0 < \overset{(4)}{\kappa}_3 < -\frac{\overset{(4)}{\kappa}_4}{2} \&\& 0 < \overset{(4)}{\kappa}_2 < \frac{4 \overset{(4)}{\kappa}_3^2 + 4 \overset{(4)}{\kappa}_3 \overset{(4)}{\kappa}_4 + \overset{(4)}{\kappa}_4^2}{36 \overset{(4)}{\kappa}_3}) \parallel ((\overset{(4)}{\kappa}_3 > -\frac{\overset{(4)}{\kappa}_4}{2} \&\& 0 < \overset{(4)}{\kappa}_2 < \frac{4 \overset{(4)}{\kappa}_3^2 + 4 \overset{(4)}{\kappa}_3 \overset{(4)}{\kappa}_4 + \overset{(4)}{\kappa}_4^2}{36 \overset{(4)}{\kappa}_3}))) \parallel$ $(\overset{(4)}{\kappa}_4 == 0 \&\& ((\overset{(4)}{\kappa}_3 < 0 \&\& (\overset{(4)}{\kappa}_2 < \frac{\overset{(4)}{\kappa}_3}{9} \parallel \overset{(4)}{\kappa}_2 > 0))) \parallel (\overset{(4)}{\kappa}_3 > 0 \&\& 0 < \overset{(4)}{\kappa}_2 < \frac{\overset{(4)}{\kappa}_3}{9}))) \parallel$ $(\overset{(4)}{\kappa}_4 > 0 \&\& ((\overset{(4)}{\kappa}_3 < -\frac{\overset{(4)}{\kappa}_4}{2} \&\& (\overset{(4)}{\kappa}_2 < \frac{4 \overset{(4)}{\kappa}_3^2 + 4 \overset{(4)}{\kappa}_3 \overset{(4)}{\kappa}_4 + \overset{(4)}{\kappa}_4^2}{36 \overset{(4)}{\kappa}_3} \parallel \overset{(4)}{\kappa}_2 > 0))) \parallel$ $(\overset{(4)}{\kappa}_3 == -\frac{\overset{(4)}{\kappa}_4}{2} \&\& (\overset{(4)}{\kappa}_2 < 0 \parallel \overset{(4)}{\kappa}_2 > 0)) \parallel (-\frac{\overset{(4)}{\kappa}_4}{2} < \overset{(4)}{\kappa}_3 < 0 \&\& (\overset{(4)}{\kappa}_2 < \frac{4 \overset{(4)}{\kappa}_3^2 + 4 \overset{(4)}{\kappa}_3 \overset{(4)}{\kappa}_4 + \overset{(4)}{\kappa}_4^2}{36 \overset{(4)}{\kappa}_3} \parallel \overset{(4)}{\kappa}_2 > 0)) \parallel$ $(\overset{(4)}{\kappa}_3 == 0 \&\& \overset{(4)}{\kappa}_2 > 0) \parallel ((\overset{(4)}{\kappa}_3 > 0 \&\& 0 < \overset{(4)}{\kappa}_2 < \frac{4 \overset{(4)}{\kappa}_3^2 + 4 \overset{(4)}{\kappa}_3 \overset{(4)}{\kappa}_4 + \overset{(4)}{\kappa}_4^2}{36 \overset{(4)}{\kappa}_3})))$	